**Radix Master v1.1 — STM32 STEM Educational Gaming Console** *A hardware-based learning system featuring 12 interactive STEM games built on STM32 Blue Pill*

Industrial STM32 Project: The Educational Gamified Logic Console

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|---|---|---|
| lood Match | Determining safe blood transfusions based on antigens. | |
| **System** | Self-Test | Hardware diagnostic for switches, encoder, and buzzer. |
| | RTC Setup | Manual adjustment of the Real-Time Clock. |
| | Return | Exit to Dashboard. |

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**4. Components & Quantity**

| **Component** | **Quantity** | **Purpose** |
|---|---|---|
| STM32F103C8T6 (Blue Pill) | 1 | Main System Controller (72MHz ARM Cortex-M3). |
| 20x4 I2C LCD | 1 | Main UI for labels, scrolling text, and feedback. |
| TM1637 4-Digit Display | 2 | High-visibility Score and Timer displays. |
| EC11 Rotary Encoder | 1 | Menu navigation and value adjustment. |
| 8-Way DIP Switch Array | 1 | 8-bit binary and logic input. |
| Active/Passive Buzzer | 1 | Auditory feedback for success/error. |
| 32.768kHz Crystal | 1 | LSE Clock source for high-precision RTC. |
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**STM32 Pin Mapping Table**

This table shows exactly where every wire from your components must land on the **STM32 Blue Pill**.

| Component | Pin Label | STM32 Pin | Logic Type |
|---|---|---|---|
| **8-Bit Switch S0** | Input 0 | **PA0** | Digital Input (Pull-up) |
| **8-Bit Switch S1** | Input 1 | **PA1** | Digital Input (Pull-up) |
| **8-Bit Switch S2** | Input 2 | **PA2** | Digital Input (Pull-up) |
| **8-Bit Switch S3** | Input 3 | **PA3** | Digital Input (Pull-up) |
| **8-Bit Switch S4** | Input 4 | **PA4** | Digital Input (Pull-up) |
| **8-Bit Switch S5** | Input 5 | **PA5** | Digital Input (Pull-up) |
| **8-Bit Switch S6** | Input 6 | **PA6** | Digital Input (Pull-up) |
| **8-Bit Switch S7** | Input 7 | **PA7** | Digital Input (Pull-up) |
| **Rotary Encoder** | A (CLK) | **PB13** | Interrupt / Input |
| **Rotary Encoder** | B (DT) | **PB12** | Input |
| **Rotary Encoder** | SW (Button) | **PB14** | Input (Pull-up) |
| **Score TM1637** | CLK | **PB10** | Serial Clock |
| **Score TM1637** | DIO | **PB11** | Serial Data |
| **Timer TM1637** | CLK | **PB0** | Serial Clock |
| **Timer TM1637** | DIO | **PB1** | Serial Data |
| **I2C 20x4 LCD** | SCL | **PB6** | I2C Clock |
| **I2C 20x4 LCD** | SDA | **PB7** | I2C Data |
| **Buzzer (Active)** | Positive (+) | **PA15** | Digital Output |

**Project Description**

Radix Master v1.1 is a hardware-based educational gaming system powered by an STM32 microcontroller. It integrates binary logic, mathematics, electronics, chemistry, and biology into a single interactive console.

Students use:

- 8-bit physical switches
- Rotary encoder navigation
- Dual 7-segment displays
- 20x4 LCD screen
- Real-time timer (RTC)

Each game challenges students to solve problems using binary logic and switch patterns, reinforcing computational thinking and scientific reasoning.

The system contains 12+ educational games including:

- Binary Blitz
- Radix Roulette
- Bitwise Logic
- Prime Hunter
- Morse Trainer
- Electron Shells
- Periodic Table
- Codon Explorer
- Gene Expression
- Blood Match

It is fully open-source and customizable.

1. **Digital Logic & Number Systems**

| Module | What it is | How to use it | Concepts Learned | Gamification Used |
|---|---|---|---|---|
| **Binary Blitz** | A rapid-fire decimal to binary translator. | Convert the decimal "Target" on the LCD into 8-bit switch positions. | Binary weights ($2^n$), Bit-positioning. | **Time Pressure:** 30s limit forces "Instant Recall" instead of manual counting. |
| **Radix Roulette** | A multi-base conversion emulator. | Convert targets appearing in **Hex (0x)**, **Octal (0)**, or **Dec** into Binary. | Cross-radix translation, Hexadecimal notation. | **Variable Reward:** Higher points for complex Hex-to-Bin conversions. |
| **Bitwise Logic** | A physical Truth Table simulator. | Given two 8-bit values and an operator (XOR, NAND, etc.), set the switches to the result. | Logic Gates (AND/OR/XOR), Masking, Boolean Algebra. | **Visual Feedback:** LCD shows "MATCH!" only when the logic gate result is perfect. |

Physical & Life Sciences (The Emulators)

| Module | What it is | How to use it | Concepts Learned | Gamification Used |
|---|---|---|---|---|
| **Electron Shells** | An atomic stability simulator. | Set the switches to match the | The Octet Rule (2, 8, 8), Atomic Numbers (Z), Valency. | **Status Simulation:** Shows "IONIC |

| Module | What it is | How to use it | Concepts Learned | Gamification Used |
|---|---|---|---|---|
| | | required valence electrons for a specific element. | | STATE" vs. "STABLE ATOM" based on input. |
| **Codon Explorer** | A Base-4 DNA-to-Protein translator. | Map 2-bit pairs (S0-S1, etc.) to A, C, G, U bases to build an Amino Acid. | Genetic Code, mRNA triplet codons, Base-4 logic. | **High Stakes:** 150-point reward for correctly "Synthesisng " a complex Protein. |
| **Gene Expression** | A phenotype activation emulator. | Match a 8-bit "Regulatory Mask" to express a trait like "Blue Eyes" or "Immunity." | Transcription Factors, Bit masking, Gene Regulation. | **Scrolling HUD:** Uses a scrolling "Phenotype" display to mimic a sequencing lab. |
| **Blood Match** | An Immuno-Logic Truth Table. | Use S0, S1, S2 as A, B, Rh antigens to match a patient's blood type. | Immunology, Antigens/Antibodies, Biological compatibility. | **Risk Assessment:** Displays "!!DANGER!!" and triggers the buzzer on a mismatch. |

3. Communications & Math

| Module | What it is | How to use it | Concepts Learned | Gamification Used |
|---|---|---|---|---|
| **Morse Trainer** | A telegraphy hardware emulator. | Use Switch 0 (S0) as a physical key to input the Morse code for a target letter. | International Morse Code, Timing (Dits vs. Dahs). | **Live Progress Bar:** LCD shows a real-time charging bar to distinguish short vs. long presses. |
| **Prime Hunter** | A numerical classification game. | Look at 8 random numbers and toggle the switches only for the **Prime** ones. | Prime vs. Composite numbers, Factorization. | **Reset Gate:** Forces the user to reset all switches to "0" before the next round to ensure focus. |

### 🕹️ User Interaction & Gameplay Flow

The **Radix Console** is designed for high-tactile engagement, moving away from a mouse and keyboard to a dedicated industrial interface. Here is the step-by-step player journey:

#### 1. System Boot & Navigation

Upon power-up, the player is greeted by the **Dashboard**, displaying the real-time clock and date. To enter the learning environment, the player clicks the **Rotary Encoder**. By rotating the dial, they scroll through the 12+ educational emulators on the 20x4 LCD. A second click selects the desired module.

#### 2. The "Ready Gate" Mechanism

To ensure fairness and "Reset Thinking," most games (like *Prime Hunter* or *Bitwise Logic*) will not start until the player toggles all **8 hardware switches** to the '0' (OFF) position. This physical "Zeroing" acts as a mental reset before the timer begins.

**3. Real-Time Problem Solving**

Once the game starts:

- **The Challenge:** The top row of the LCD displays a target (e.g., a Decimal number, a Chemical element, or a DNA Codon).
- **The Input:** The player manipulates the **8-bit switch array**. In **Binary Blitz**, they map values to binary weights ($2^n$); in **Codon Explorer**, they treat pairs of switches as A, C, G, or U bases.
- **The Feedback:** The left **TM1637 display** shows the live score, while the right display counts down the remaining seconds.

**4. Success & Evolution**

When the player matches the target logic, the **Active Buzzer** provides an immediate "Success Beep," and the score increments. In the **Gene Expression** emulator, the LCD might scroll a new "Phenotype" to express, forcing the player to quickly recalculate their bitmask. If the timer hits zero, the final score is flashed across the LCD, and the player is returned to the menu to try a different challenge.